

Introduction to Algorithms

Topic 0 : Course Information

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Fall Semester 2026

- ▶ **Lecture Time and Room**

- ▶ Wednesday, 14:00–15:35
- ▶ Friday, 09:45–11:20
- ▶ Room GT-B212

- ▶ **Credit Hours:** 60 (Theory) + 30 (Experiment), 3.5 credits

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▶ Textbook and Recommended References

- ▶ **Textbook:** 《Introduction to Algorithms》, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein.
中文翻译版: 《算法导论》, 机械工业出版社. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein 著. 潘金贵, 顾铁成, 李成法, 叶懋 译
- ▶ **Main Reference:** 《Algorithm Design》影印版 (中文名: 算法设计), 清华大学出版社. Jon Kleinberg, Eva Tardos 著

Course Objectives

- ▶ **A foundational course in computer science.**
 - ▶ Study the design and analysis of algorithms.
 - ▶ Emphasis on searching, sorting, and combinatorial algorithms.
 - ▶ Analyze worst-case and average-case time and space complexity.
 - ▶ Develop practical skills in implementing efficient algorithms.
- ▶ **Prerequisites**
 - ▶ 程序设计, 数据结构, 高等数学, 离散数学

Course Outline

- ▶ Basic Concepts
- ▶ Asymptotic Notation and Recurrences
- ▶ Comparison-based Sorting Algorithms
 - ▶ insertion sort, Shellsort, quicksort
- ▶ Linear-time Sorting and Order Statistics
 - ▶ counting sort, radix sort, bucket sort, selection
- ▶ Advanced Data Structures
 - ▶ binary search trees, red-black trees
- ▶ Algorithm Design Techniques
 - ▶ dynamic programming, greedy methods, divide-and-conquer
- ▶ Graph Algorithms
 - ▶ DFS, BFS, minimum spanning trees, shortest paths
- ▶ String Matching
 - ▶ brute force, KMP, Shift-Or, BM, BMH, Quick Search, KR
- ▶ NP-Completeness and Approximation Algorithms

Algorithms and AI: A Two-Way Interaction

Algorithm for AI

Classical algorithms provide the computational foundation for:

- ▶ model training and inference;
- ▶ retrieval and data processing;
- ▶ system scheduling and optimization.

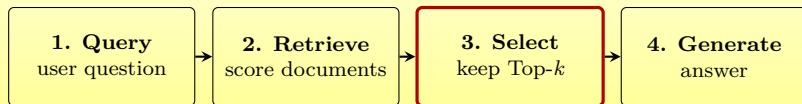
AI for Algorithms

AI can augment classical algorithms through:

- ▶ learned predictions;
- ▶ algorithm design;
- ▶ formal verification;
- ▶ constraint solving.

Algorithm for AI: RAG Retrieval

Retrieval-augmented generation (RAG) sends only the most relevant documents to the model.



A Concrete Example ($k = 2$; illustrative scores)

Query: When and where is Wednesday's Algorithms class?

D1 (0.95): Wednesday: 14:00–15:35. D2 (0.91): Room: GT-B212.

D3 (0.52): Friday: 09:45–11:20.

D4 (0.30): Recitation: Tuesday.

D5 (0.18): TA office hours.

Selected Top-2: D1, D2.

Answer: Wednesday's class is from 14:00 to 15:35 in GT-B212.

Algorithmic Question

Among n scored candidates, full sorting costs $O(n \log n)$. A size- k min-heap keeps only the best k in $O(n \log k)$ time and $O(k)$ extra space.

AI for Algorithms: Four Directions

- ▶ **Learning-augmented algorithms:** use predictions in caching, scheduling, search, and pruning, with safeguards against prediction errors.
- ▶ **LLM-assisted algorithm design:** generate and refine candidate algorithms; analyze complexity and diagnose bottlenecks.
- ▶ **LLM-assisted formal verification:** generate specifications and invariants for proof tools to check.
- ▶ **LLM-assisted constraint solving:** encode requirements as SAT/SMT/CP/MIP constraints and refine them using solver feedback.

Key Principle

AI supplies predictions, candidate designs, and formal encodings; algorithmic safeguards, proof tools, and solvers provide robustness, correctness, and feasibility guarantees.

► Assignments and Experiments (25%)

- Assignments: 10 homework sets, assigned approximately weekly, with **firm** deadlines
- Six experiments (tentative schedule):

1. 排序算法及性能比较

Tentative date: 2026-09-27

2. 高级数据结构: 红黑树、数据结构扩充、二项堆等

Tentative date: 2026-10-11

3. 动态规划: LCS、矩阵链乘法、最优二叉搜索树等

Tentative date: 2026-10-25

4. 贪心算法: 区间覆盖、 K 进制编码、活动选择、背包问题等

Tentative date: 2026-11-08

5. 图算法: 所有点对最短路径、强连通分量等

Tentative date: 2026-11-22

6. 字符串匹配: KMP、BM、KR、Quick Search 等

Tentative date: 2026-12-06

- ▶ Assignments and Experiments: 25%
- ▶ Midterm Examination: 20% (tentative date: 2026.10.31)
- ▶ Final Examination: 40% (during the examination week)
- ▶ Attendance and Participation: 15%
 - ▶ Attendance and in-class quizzes: 10%.
 - ▶ Active students (e.g., interacting with instructors) will win the other 5 points.

Grading Policy

- ▶ The instructor reserves the right to make adjustments to these weights based on his a posteriori evaluation of the relative difficulty of the exams and homework.
- ▶ Each problem will be graded 80% for correctness and 20% for style and clarity.
- ▶ **Final Grade** $W = \frac{W_1 + W_2}{2}$, W_1 is the final weighted score (Assignments and Experiments + Attendance + Midterm + Final) and $W_2 = 100 \times \frac{W_1}{AverageTopFive}$, where *AverageTopFive* is the average of W_1 of the best five students in the class. For example, if your $W_1 = 70$, and $AverageTopFive = 90$, then your $W = \frac{70 + 70 \cdot 100 / 90}{2} \simeq 73.89$.
- ▶ **No plagiarism will be tolerated**

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- ▶ 王天竹, wtz2333@mail.ustc.edu.cn
- ▶ Weekly Recitation: Tuesdays, 15:00–16:00 (tentative; to be confirmed with students).
- ▶ Weekly Office Hours: each TA will offer weekly office hours; the schedule will be announced after coordination with the TA team.

► Course Homepage:

<https://2026-ustc-algorithm.github.io/>

Course materials and announcements will be released progressively on the course homepage and in the QQ group.

- course information and schedule
- released lecture slides
- assignments and other course materials

► Online Judge:

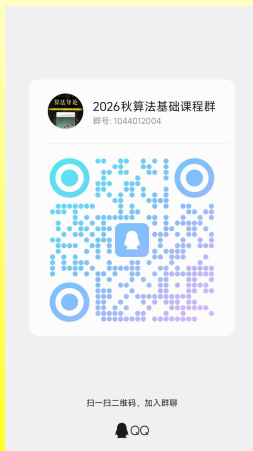
https://hydro.ac/d/USTC_Algorithm_2026/

- Programming experiments will be submitted and evaluated here.

► BB Platform: <https://www.bb.ustc.edu.cn/>

- Upload your assignments here
- Writing assignments with Latex (highly recommended), MS Word, or just taking a photo of your answers on a paper.

Course announcements will also be posted in the QQ group.



Wish You Enjoy This Course!